

Object Oriented Programming | Keywords, Data Types, Identifiers

Two Main Types of Data in Java

1. Primitive Data Types

Primitive data types are the **basic data types** in Java. They store **simple values** directly.

Examples: `int`, `float`, `char`, `boolean`

2. Classes

A class is a **user-defined data type** that contains **data and methods**. Objects are created from classes.

Example: `Student`, `Employee`

Short Difference

Primitive Data Types	Classes
Store simple values.	Define objects.
Built into Java.	Created by the programmer.
Example: <code>int</code> , <code>char</code>	Example: <code>Student</code> , <code>Employee</code>

Differences Between `7`, `'7'` and `"7"`

Constant	Type	Meaning
<code>7</code>	Integer (<code>int</code>)	Represents the number 7
<code>'7'</code>	Character (<code>char</code>)	Represents the single character 7
<code>"7"</code>	String (<code>String</code>)	Represents the text/string "7"

Example

```
int a = 7;
char b = '7';
String c = "7";
```

Identifiers in Java

Identifiers are the **names given to classes, variables, and methods** in Java.

Rules of Identifiers

- An identifier can contain **letters** (A–Z, a–z), **digits** (0–9), `$`, and `_`.
- An identifier **must begin with a letter**, `$`, or `_`.
- An identifier **cannot start with a digit**.
- A **Java keyword** cannot be used as an identifier.

- Identifiers are **case-sensitive**.

Example of Case Sensitivity

```
area
Area
AREA
```

Legal Identifiers

```
age
$salary
_value
__1_value
```

Illegal Identifiers

```
123abc      // Cannot start with a digit
-salary     // '-' is not allowed
student name // Space is not allowed
class       // 'class' is a keyword
```

Easy Way to Remember

An identifier is simply a name used for a class, variable, or method.

Find Out the Legal Identifiers

Legal Identifiers

- \$2
- area
- Min_number
- aaa
- sales_tax
- _circleArea
- a2b
- Box100width
- ab1234\$\$
- ComputeArea

Illegal Identifiers ❌

- `2A` → Cannot start with a digit.
- `d+4` → `+` is not allowed.
- `Max-Number` → `-` is not allowed.
- `/$directory` → `/` is not allowed.

Final Answer

Legal: `$2`, `area`, `Min_number`, `aaa`, `sales_tax`, `_circleArea`, `a2b`, `Box100width`, `ab1234$$`, `ComputeArea`

Illegal: `2A`, `d+4`, `Max-Number`, `/$directory`